

Research Movement Report

Facial movement classification and recorded action evidence.



MEEI FACIAL-MOVEMENT CLASSIFICATION SCORE

48 / 100

2 points below the fixed cutpoint of 50.

What this number represents

This is the average of three model outputs on the MEEI facial-palsy-versus-healthy-control classification scale. Higher scores lean toward the facial-palsy class; lower scores lean toward the healthy-control class.

CURRENT OUTPUT Healthy-control class

2 points below the cutpoint



How the model formed the score

Recorded actions → MediaPipe facial geometry → shared encoder → MEEI classification score.

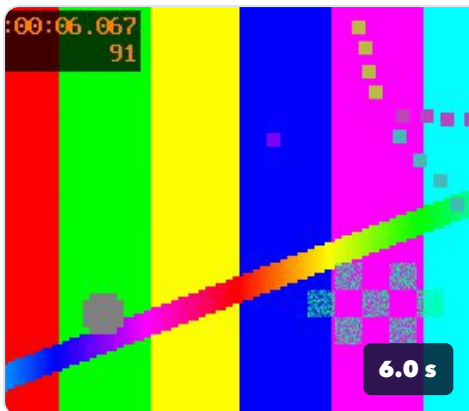


Recorded action evidence

Each context image is taken at the registered midpoint of its three-second hold. It is recorded context, not a frame selected by the model.

🛡️ **Measured movement observation – not a cause of the model score or a clinical severity grade.**

0.010 means 1% of the distance between the eyes. These descriptive geometry values have no clinical normal range or severity meaning.



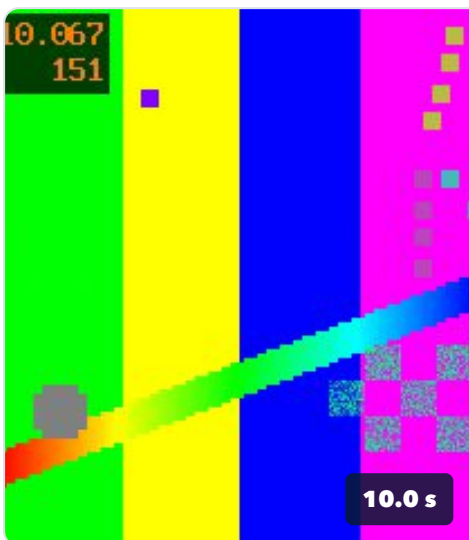
Eyebrow raise

Left-right brow-height difference

0.010 × inter-eye distance

Brow-height change from rest

0.020 × inter-eye distance



Gentle eye closure

Left-right eye-opening difference

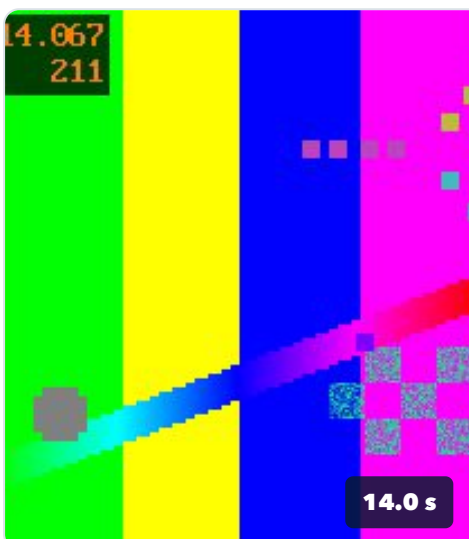
0.010 × inter-eye distance

Eye opening during the hold

0.020 × inter-eye distance

Eye-opening change from rest

0.030 × inter-eye distance



Tight eye squeeze

Left-right eye-opening difference

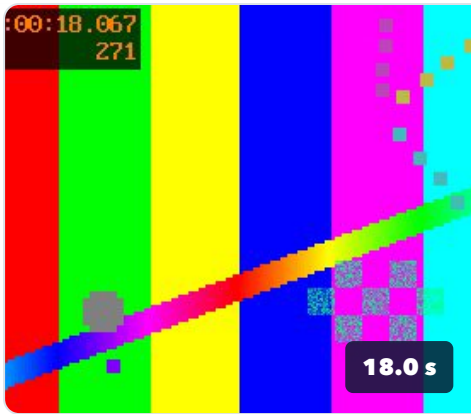
0.010 × inter-eye distance

Eye opening during the hold

0.020 × inter-eye distance

Eye-opening change from rest

0.030 × inter-eye distance



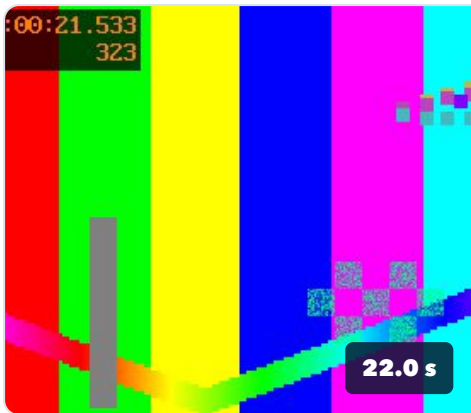
Relaxed smile

Left-right mouth-corner height difference

0.010 × inter-eye distance

Mouth-corner movement from rest

0.020 × inter-eye distance



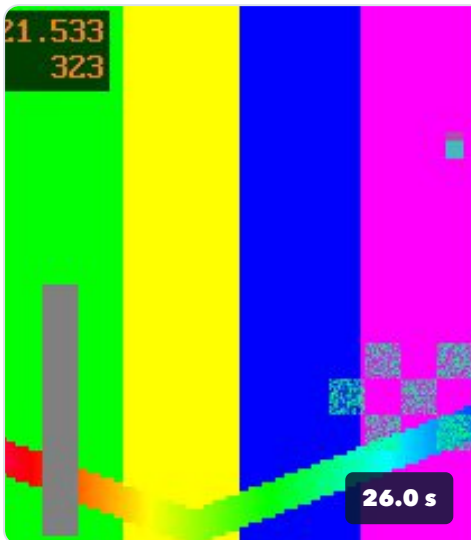
Lip pucker

Left-right mouth-corner width difference

0.010 × inter-eye distance

Mouth-width change from rest

0.020 × inter-eye distance



Show lower teeth

Left-right mouth-corner height difference

0.010 × inter-eye distance

Lower-lip movement from rest

0.020 × inter-eye distance

Mouth-opening change from rest

0.030 × inter-eye distance

Recording coverage

Recorded steps included in this score

Neutral baseline + all 6 active movements

Face tracking coverage

26-31 usable of 32 checkpoints per movement

Optional Step 8

Not part of this session



All 7 recorded steps in this session were used: one neutral baseline and 6 active movements.

Each active movement is checked at 32 evenly spaced time points; the range above shows how many had usable face tracking. The neutral recording provides the resting baseline used for movement-change measurements.



Interpretation limits

This research model has not been clinically validated for Mayo patients and does not provide diagnosis, etiology, affected-side identification, or treatment guidance. Scores close to the cutpoint may be sensitive to recording and population differences. House-Brackmann, Sunnybrook, eFACE, and FaCE require separate clinician or patient assessment and are not generated in this report.